

Sample Script 2

```

• import torch
• import torchvision.transforms as T
• from PIL import Image
•
• # Load the diffusion model
• model = torch.load('diffusion_model.pt', map_
location='cpu')
•
• # Load the input image
• input_image = Image.open('input_image.jpg')
•
• # Preprocess the input image
• preprocess = T.Compose([
•     T.Resize(256),
•     T.CenterCrop(256),
•     T.ToTensor(),
• ])
• input_tensor = preprocess(input_image).unsqueeze(0)
•
• # Generate an output image
• with torch.no_grad():
•     output_tensor = model.sample(input_tensor, diffu-
sion_steps=1000)
•
• # Postprocess the output image
• output_image = T.functional.to_pil_image(output_
tensor.squeeze())
•
• # Save the output image
• output_image.save('output_image.jpg')

```

This script loads a pre-trained stable diffusion model from a file, loads an input image, preprocesses it, generates an output image using stable diffusion with 1000 diffusion steps, post-processes the output image, and saves it to a file.

Note that this method of the stable diffusion model takes an additional argument `diffusion_steps` which specifies the number of diffusion steps to perform. A larger value results in a more accurate but slower diffusion process. The script sets it to 1000, but you can adjust it to trade-off between speed and accuracy.